During Stage 2, both model checkpointing and test set evaluations occurred every 10 training steps. The checkpoint yielding the highest test score was selected as the final Stage 2 model (θ_2).

These tailored configurations allowed for progressive and adaptive learning, ensuring that Time-R1 developed foundational understanding before advancing to more complex predictive tasks.

B Dataset Construction and Details

This appendix provides further details on the datasets used for training and evaluating Time-R1, supplementing the descriptions in Sections 3.2.1 and 3.2.2.

B.1 New York Times (NYT) Corpus Curation

The primary data source for our research is a corpus constructed from New York Times articles, utilizing publicly available information accessed via the NYT Archive API[2]. For each article, we extracted key fields including the headline, abstract, publication date, and the "news desk" (thematic section).

We collected over 200,000 English-language NYT articles, with publication dates spanning from January 2016 to February 2025. To ensure the relevance of the articles to common temporal reasoning scenarios and current events, we selectively curated content from the following news desks: "Politics", "National", "Washington", "U.S.", "Business", "SundayBusiness", "RealEstate", "Foreign", "World", "Metro", "Science", "Health", "Climate", "Opinion", and "OpEd". Other news desks were excluded as they were found to reference current events less frequently.

This extensive NYT corpus was utilized for several distinct purposes within our framework:

- **Stage 1 (Comprehension) Training Data:** Articles published from January 2016 to December 2023 were used to train the foundational temporal understanding capabilities of Time-R1 (see Section 3.2.1 for Stage 1 details).
- **Stage 2 (Prediction) Real News Training Data:** A subset of articles from January 2024 to July 2024 served as real-world news data for the initial phase of Stage 2 training.
- **Stage 2 (Prediction) Real News Test Data:** Articles from August 2024 to February 2025 were held out and used as the real-news test set ($\mathcal{D}_{\text{test}}^{(2)}$) for evaluating future event prediction performance.

In our task formulations, an event E is typically represented by its headline h and abstract a, i.e., $E = (h, a)$.

B.2 Synthetic Data Generation for Future Event Prediction Training

To train Time-R1 for predicting events in future months (specifically, August 2024 to February 2025) without encountering data leakage from the real-news test period, we employed a data synthesis strategy as detailed in Section 3.2.2. This process utilized the DeepSeek-V3 model with a knowledge cutoff in July 2024.

The methodology for generating synthetic news articles was as follows:

- **Targeted News Desk Distribution:** The generation aimed to reflect a historical distribution of articles across various news desks, based on NYT data prior to 2024. The primary target desk distribution used to guide generation proportions was:

  Foreign: 20.8%; Business: 16.5%; OpEd: 14.2%; National: 10.9%; Washington: 9.6%; Metro: 8.6%; Politics: 5.5%; Science: 4.6%.
- **Few-Shot Prompting Strategy:** To generate content for a specific target future month (between August 2024 and February 2025) and a designated news desk, the DeepSeek-V3 model was prompted using a few-shot learning approach. Each prompt contained three real news headlines and abstracts from the *same* news desk, randomly sampled from articles published between May 2024 and July 2024.

[2] `https://developer.nytimes.com/docs/archive-product/1/overview`

- **Generation Task:** For each such prompt, DeepSeek-V3 was instructed to generate six distinct synthetic news items (each comprising a headline and an abstract) relevant to the specified future month and news desk, learning from the style and content of the provided examples.
- **Output Distribution:** The selection and aggregation of these generated articles were managed so that the overall proportion of news items per desk for each future month in the synthetic training set ($\mathcal{D}^{(2)}_{\text{train}}$) approximately mirrored the historical desk distribution detailed above.

This synthetic dataset provided the necessary training signals for the model to learn to predict events beyond its real-data cutoff while strictly ensuring no overlap with the real-news test data from the same period. The volume of this synthetic data for August 2024 - February 2025 was about half that of the real news data used for January 2024 - July 2024 in the Stage 2 training.

C Detailed Stage 1 Learning Curves and Analysis

This section provides a more detailed look at the learning dynamics during Stage 1 (Comprehension), complementing the summarized performance presented in Table 1 of Section 4.4.1. We present the training curves for all four fundamental temporal subtasks—Timestamp Inference, Time-Difference Estimation, Event Ordering, and Masked Time Entity Completion—specifically focusing on their progression throughout Phase 2 and Phase 3 of our dynamic reward curriculum (see Section 3.3.3 for details on the curriculum phases). Additionally, we illustrate the evolution of the inconsistency penalty factor (P_{incon}) for the Time-Difference Estimation and Event Ordering tasks during Phase 2, highlighting the model's improving adherence to logical and mathematical consistency.

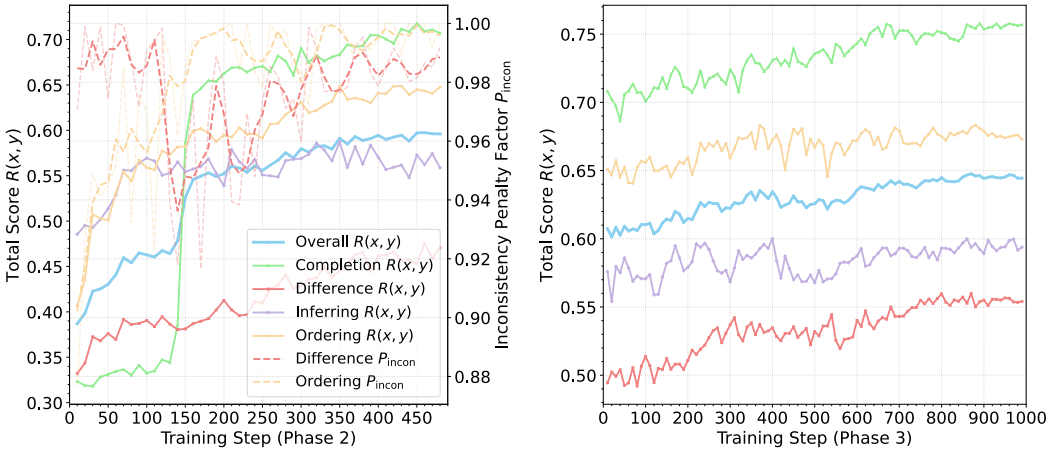

Figure 6: Learning curves for Stage 1 subtasks during (Left) Phase 2 and (Right) Phase 3 of the dynamic reward curriculum. The left plot also shows the Inconsistency Penalty Factor (P_{incon}) for Time-Difference Estimation and Event Ordering tasks on the right y-axis during Phase 2.

The learning curves depicted in Figure 6 offer several key insights into the effectiveness of our methodology. Firstly, the steady increase and eventual convergence of the total scores ($R(x, y)$)) across all subtasks in both Phase 2 and Phase 3 underscore the benefits of our dynamic reward design and curriculum learning strategy. This carefully structured approach enables the model to progressively master complex temporal logic, gradually adapting from more lenient to stricter evaluation criteria. As noted in Section 4.4.1, this robust Stage 1 performance allows our 3B Time-R1 model to surpass numerous baseline models, many of which are ten to over two hundred times larger in parameter count (Table 1). Such strong foundational capabilities in temporal comprehension are crucial and deliberately engineered to provide a solid grounding for the subsequent, more demanding future-oriented tasks in Stage 2 (Prediction) and Stage 3 (Generation).

Secondly, the trends observed for the inconsistency penalty factors (P_{incon}) for the Time-Difference Estimation and Event Ordering tasks during Phase 2 (left plot of Figure 6, dashed lines) are particularly

revealing. The increasing values of P_{incon} (approaching 1.0) indicate that the model is effectively learning to minimize inconsistencies in its responses. For instance, in Time-Difference Estimation, it learns to ensure that the explicitly stated time difference aligns with the difference calculated from its inferred dates for the two events. Similarly, for Event Ordering, the model becomes better at ensuring the stated order of events is consistent with the chronological sequence implied by its inferred dates for those events. This demonstrates that the penalty mechanisms detailed in Section 3.3.2 successfully guide the model not just towards task-specific accuracy but also towards generating responses that are logically coherent and mathematically sound, a critical aspect of true temporal understanding. By the commencement of Phase 3, these consistency factors are generally high, allowing the training to focus further on refining accuracy under strict evaluation.

Overall, these detailed learning dynamics from Stage 1 highlight the efficacy of our curriculum in building both accurate and logically consistent temporal reasoning, providing the essential groundwork for Time-R1's advanced capabilities in navigating future temporal challenges.

D Further Discussion on Implementation Settings

This appendix elaborates on specific implementation settings, focusing on the impact of the KL loss coefficient on model response length and the overall stability of our training framework with respect to various hyperparameter changes. These details supplement the primary configurations presented in Table 4.

D.1 Impact of KL Loss Coefficient on Response Length

The Group Relative Policy Optimization (GRPO) objective function, as defined in Equation (3), incorporates a KL divergence term $\mathbb{D}_{KL}[\pi_\theta(\cdot|x)||\pi_{ref}(\cdot|x)]$ scaled by a coefficient β. This term penalizes deviations of the current policy π_θ from a reference policy π_{ref}, encouraging smoother and more stable policy updates. The magnitude of β directly influences the strength of this regularization.

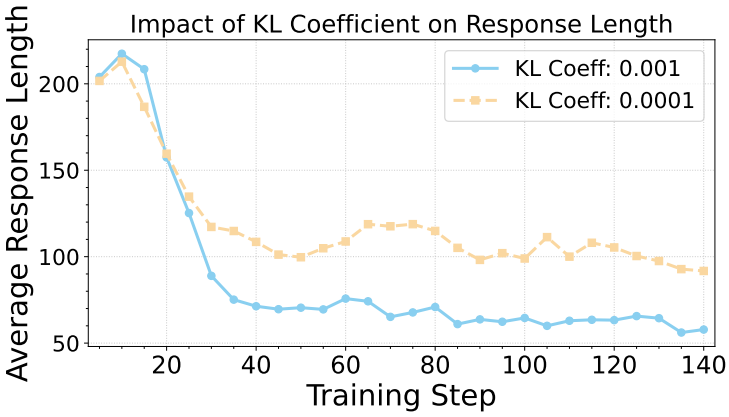

Figure 7: Impact of different KL loss coefficients (β) on the average response length during training. A lower coefficient (0.0001) leads to longer average responses compared to the default setting (0.001).

As illustrated in Figure 7, a lower KL coefficient (e.g., $\beta = 0.0001$ compared to our default $\beta = 0.001$) reduces the penalty for deviating from the reference policy. This allows the model greater freedom to explore diverse generation strategies during training. A noticeable consequence of this increased exploration with a lower β is an increase in the average length of the generated responses. However, our experiments indicated that while the response lengths varied, the overall performance scores on the test sets remained largely comparable across these KL coefficient settings. This suggests that while the KL coefficient can influence stylistic aspects of the generation, such as verbosity, the core temporal reasoning capabilities learned by the model are robust within this range of β values.